

A support face is the convex hull of the maximizing generators

MathlibAnnex.convexHull_maxSlice_eq_supportFace

theorem

Identifies the points of a convex hull on a supporting level by discarding generators with zero weight.

Statement

Let K be a nonempty compact subset of a finite-dimensional real normed space and let ℓ be a continuous linear functional. Set

$$m = \max_{x \in K} \ell(x), \quad K_\ell = \{x \in K : \ell(x) = m\}.$$

Then

$$\text{conv}(K_\ell) = \{z \in \text{conv}(K) : \ell(z) = m\}.$$

The right side is the support face at the maximum of ℓ .

Assumptions

The ambient space is a finite-dimensional real normed vector space. The set K is nonempty and compact, and ℓ is continuous and real-linear. The set K need not be convex; ℓ may be zero.

Conclusion

The entire support face is generated by the original points of K at which ℓ attains its maximum. In particular, one need not introduce new generators from the interior of the convex hull.

Notes

Continuity and compactness give a maximizer in K , so K_ℓ is nonempty. It is the intersection of K with the closed level set $\ell^{-1}(\{m\})$, hence compact. These two facts make the maximum slice eligible for another refinement; membership in that refined set is precisely membership in K together with equality to the maximum.

Proof route

One inclusion follows from convexity of a level set. For the reverse inclusion, a convex representation at the maximum has zero total gap, forcing every positive-weight generator to lie at that maximum.

▼ Proof steps

1. The maximum exists because K is nonempty compact and ℓ is continuous. Put $H = \{x : \ell(x) = m\}$. Linearity makes H convex, and $K_\ell \subseteq K \cap H$. Hence

$$\text{conv}(K_\ell) \subseteq \text{conv}(K) \cap H.$$

Here the first containment uses both monotonicity of the convex hull and convexity of H . It remains valid when $\ell = 0$.

2. Conversely, let $z \in \text{conv}(K)$ satisfy $\ell(z) = m$. Choose a finite convex representation $z = \sum_i t_i x_i$ with $x_i \in K$, $t_i \geq 0$ and $\sum_i t_i = 1$. Each gap $m - \ell(x_i)$ is nonnegative, and linearity gives

$$\sum_i t_i (m - \ell(x_i)) = m \sum_i t_i - \ell\left(\sum_i t_i x_i\right) = m - \ell(z) = 0.$$

A finite sum of nonnegative terms is zero only if every term is zero. Therefore $t_i > 0$ implies $m - \ell(x_i) = 0$, so $x_i \in K_\ell$.

3. Remove the zero-weight terms. The remaining weights still sum to 1, their weighted sum is still z , and every remaining generator belongs to K_ℓ . Thus $z \in \text{conv}(K_\ell)$, proving the reverse inclusion and the equality.

Main citations

- Exact declaration and its source — `MathlibAnnex.convexHull_maxSlice_eq_supportFace`
- A maximizing point on a nonempty compact set — `MathlibAnnex.NonemptyCompacts.maximizer`
- The maximizing point belongs to the set — `MathlibAnnex.NonemptyCompacts.maximizer_mem`
- Every value is bounded by the maximum — `MathlibAnnex.NonemptyCompacts.le_maximizer`
- The maximum value — `MathlibAnnex.NonemptyCompacts.maxValue`
- The raw maximum slice — `MathlibAnnex.NonemptyCompacts.maxSlice`
- The maximizer belongs to its maximum slice —
`MathlibAnnex.NonemptyCompacts.maximizer_mem_maxSlice`
- The maximum slice is nonempty — `MathlibAnnex.NonemptyCompacts.maxSlice_nonempty`
- The maximum slice is compact — `MathlibAnnex.NonemptyCompacts.isCompact_maxSlice`
- The nonempty compact refinement — `MathlibAnnex.NonemptyCompacts.refine`
- Membership in the refinement — `MathlibAnnex.NonemptyCompacts.mem_refine`
- The support face in the convex hull — `MathlibAnnex.supportFace`

Lean source signature (exact)

```
theorem convexHull_maxSlice_eq_supportFace [FiniteDimensional ℝ E]
  (K : TopologicalSpace.NonemptyCompacts E) (ℓ : E →L[ℝ] ℝ) :
  convexHull ℝ (maxSlice K ℓ) = supportFace K ℓ
```

In the source

```
{E : Type u} [NormedAddCommGroup E]
[NormedSpace ℝ E]
[FiniteDimensional ℝ E]
```

Mathematical meaning

The finite-dimensional real normed ambient space.

In the source	Mathematical meaning
<code>(K : TopologicalSpace.NonemptyCompacts E)</code>	The set K with its nonemptiness and compactness; convexity is not included.
<code>(ℓ : E →L[ℝ] ℝ)</code>	The continuous real-linear functional ℓ , possibly zero. In this Card $m = \max_{x \in K} \ell(x)$.
<code>maxSlice K ℓ</code>	The original maximum slice $K_\ell = \{x \in K : \ell(x) = m\}$, as defined in the cited declaration.

In the source	Mathematical meaning
<code>supportFace K ℓ</code>	The set $\{z \in \text{conv}(K) : \ell(z) = m\}$, as defined in the cited support-face declaration.
<code>convexHull ℝ (maxSlice K ℓ) = supportFace K ℓ</code>	The conclusion $\text{conv}(K_\ell) = \{z \in \text{conv}(K) : \ell(z) = m\}$.

▼ Exact surrounding binder context (separate excerpt)

```

noncomputable section

open Set
open TopologicalSpace

namespace MathlibAnnex

universe u

variable {E : Type u} [NormedAddCommGroup E] [NormedSpace ℝ E]

```

▼ Exact content identity

Declaration: `MathlibAnnex.convexHull_maxSlice_eq_supportFace`

Accepted content SHA-256: `8b332398f902875f464ae106ef414d493cbd6508fad62cb990831c6250d058ce`

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